

kidneys, no history of neck irradiation. Patients were treated 2 Gy once-daily fractionation with a total dose of 60 Gy in a 6-week overall treatment time. Concurrently, TS-1 was given orally at a fixed dose of 80 mg/m²/day in a split-course fashion with a 2-week rest after a 2-week administration (Level 1), and with a one-week rest after a 3-week administration (Level 2). The DLT was defined as grade 4 hematological toxicities, or grade 3 or more non-hematological toxicities.

Results: DLT was not observed after 4 patients were registered at dose level 1, and then 25 patients were registered at level 2 between 2003 and 2006. In all evaluated patients, the gender was 28 in male and one in female. The primary site of the cancer was the glottis in 26 and the supraglottis in 3. Clinical stage was 28 in stage II and one in III. Of 25 patients enrolled at level 2, three patients discontinued an administration of TS-1 because of DLT (severe mucositis in two, liver dysfunction in one), and one patient was necessitated to be given a reduced dose of TS-1 after the break. No patients had a local progression during the follow-up time of 13-60 months, but only one patient developed the regional node recurrence, and one patient died from intercurrent disease.

Conclusions: Concurrent use of TS-1 with 80mg/m² in a split-course with one-week rest during the course of radiation therapy for stage II-III laryngeal cancer was considered safe and active, and better than injecting patients with anticancer drugs for quality of life (QOL).

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2564 Visual Acuity and Complications after Ruthenium-106 Plaque Brachytherapy for Medium Size Choroidal Melanoma

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Purpose/Objective(s): Review treatment outcomes for medium-size choroidal melanoma with Ruthenium-106 plaque brachytherapy.

Materials/Methods: Study population consisted of 20 patients (10 males and 10 females) treated with Ruthenium-106 (Ru-106) plaque brachytherapy between 2001 and 2006. The median age was 67 years (range, 40-87 years). All patients were evaluated by funduscopy; A & B scan ultrasonography, and fluorescein angiography. Systemic assessment included chest radiograph and abdominal ultrasound. The median tumor apical height was 3.3 mm (range, 2.3-5.25 mm) and median basal diameter of 10.5 mm (range, 7-14 mm). The prescribed tumor dose was 85 Gray (Gy) to a depth of 5 mm. Analysis of visual acuity 12 months post-irradiation was available for 17 patients and 24 months post-irradiation for 14 patients. Both pre- and post-irradiation ultrasound reports were obtained for 15 patients.

Results: The median follow-up for the 20 patients was 26.5 months (range 3-51 months). The median radiation dose was 84.02 Gy (range 81.4-88.23 Gy) to 5 mm depth, with a median treatment time of 98.15 hours (range 68-170.9 hours). The median dose to tumor apex was 148.6 Gy (range 78.17-210.0 Gy). Visual acuities $\geq 20/70$ were observed in 64.7%, and 64.3%, 12- and 24-months post-irradiation, respectively. Two patients maintained 20/20 vision and two patients had improved vision 24-months post-irradiation. There was no enucleation, post-irradiation glaucoma, or local recurrence of tumor as of most recent follow-up for our patients. Two patients developed cataracts in the treated eye at 20 and 37 months follow-up. One patient presented with partial retinal detachment and 18 months post-irradiation experienced photophobia with complete retinal detachment. One patient was found to have radiation retinopathy; however he maintained a visual acuity of 20/40 in the treated eye. Ninety-three percent of tumors regressed post-irradiation, with only 1 tumor remaining stable in size 24 months post-treatment.

Conclusions: Our experience is similar to other European centers using Ru-106 plaque brachytherapy. For medium size choroidal melanoma, Ru-106 plaque brachytherapy is an efficacious conservative treatment modality.

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2565 Preliminary Report on Ongoing Quality of Life Evaluation of Patients with Locally Advanced Head and Neck Cancer undergoing Intensity Modulated Radiotherapy (IMRT) with Chemotherapy: The Multi-disciplinary Bronx Experience

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Purpose/Objective(s): To evaluate quality of life in patient undergoing concurrent chemotherapy and IMRT for the treatment of locally advanced head and neck cancer.

Materials/Methods: Over 100 patients have been treated for locally advanced head with concurrent chemotherapy and IMRT from 2005-2008. We have evaluated quality of life using the EORTC QLQ-C30 and QLQ-H&N 35. We have thus far collected data on the first 20% of this cohort treated from 2005-2008. We have evaluated the patients based on both level of functioning and level of symptomatology. All patients were treated using IMRT with simultaneous integrated boost (SIB) with concurrent chemotherapy. Patients were treated with cisplatin, carboplatin+/- 5-FU or cetuximab. The prescription dose encompassed at least 95% of the PTV at the 95% isodose line and the hot spots were kept below 110% for all plans. All patients were treated with IMRT using Varian Eclipse treatment planning system. Patients were treated to 66-70 Gy (2.12 Gy/Fraction) to the primary PTV and 54-60 Gy (1.64-1.8 Gy/fraction) to the subclinical and high risk PTV. Questionnaires were given to patients during follow-up in radiation oncology, medical oncology and with head and neck surgeons.

Results: Preliminary analysis at a median follow-up of 15 months has shown a high level of functioning and health with low persistent symptomatology. The Global health status mean score was 72 with a standard deviation (SD) of 26. The level of functioning was high in physical (mean 88, SD 18), role (mean 81, SD 32), social (mean 92, SD 14) and cognitive (mean 79, SD 29). Analysis of symptoms showed a low level of persistent problems with the exception of dry mouth (mean 55, SD 30), sticky saliva (mean 48, SD 43) and use of nutritional supplements (mean 53, SD 51). Most patients evaluated felt well, which was reflected by a low symptoms and a low score in the felt ill category (mean 4.4, SD 12).